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Performance of 3D-printed SiC-reinforced Ti6Al4V and its high thermal conductivity brazing with SiC ceramics



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Highlights:

- First systematic investigation on brazing behavior between 3DP-SiC/TC4 and SiC ceramics.
- AgCuTi braze shows enhanced wettability on 3DP-SiC/TC4 by porous EBM structure and SiC particles.

- A TiC + Ti₅Si₃ layer ($\sim 1.0\mu\text{m}$) forms at the SiC side, enabling strong bonding and efficient thermal transport.
- Brazed joint achieves 39.5 MPa shear strength and $21.6\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ thermal conductivity at 600°C, 57% and 56% higher than TC4 joints, respectively.
- A thermal resistance model reveals that brazing reduces total interfacial thermal resistance by 83.2%.

Abstract

The development of efficient thermal management systems urgently requires reliable joining between 3D-printed SiC-reinforced Ti6Al4V composites (3DP-SiC/TC4) and SiC ceramics. This study comprehensively investigates the microstructure, physical properties and wettability of 3DP-SiC/TC4 fabricated by electron beam melting (EBM). Compared with original TC4, 3DP-SiC/TC4 exhibits a lower coefficient of thermal expansion ($8.4\times 10^{-6}\cdot\text{K}^{-1}$) and a higher thermal conductivity ($11.647\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$). High-quality brazed joints are achieved between 3DP-SiC/TC4 and SiC ceramics using AgCuTi braze. Microstructural evolution, mechanical properties and thermal characteristics are systematically investigated. A layered reaction zone preferentially forms at SiC interfaces, consisting of TiC and Ti₅Si₃ with a total thickness of $\sim 1.0\mu\text{m}$. The central region of the joint develops a mixed microstructure of Ag(s,s) and Ti-Cu intermetallic compounds, while graded Ti-Cu diffusion zones form on the 3DP-SiC/TC4 side. The heat transfer analysis demonstrates that brazing effectively reduces the interfacial thermal resistance by filling the air gaps. The optimal brazing temperature is 850°C, achieving joints with the shear strength of 39.5 ± 1.0 MPa and the thermal conductivity of $21.6\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ (tested at 600°C), representing 57% and 56% improvements over original TC4 joints respectively. This research provides innovative material joining solutions for lightweight aerospace thermal management system design.

Keywords

SiC; 3D-printed SiC-reinforced Ti6Al4V alloy; Brazing; Thermal conductivity

1. Introduction

With rapid advancements in aerospace and high-power electronic devices, thermal management systems increasingly face dual challenges of weight reduction and efficient heat dissipation [1], [2], [3], [4]. Ti-6Al-4V(TC4) titanium alloy has been considered an ideal metallic material for lightweight heat dissipation components due to its excellent specific strength, corrosion resistance, and manufacturability [5], [6], [7], [8], [9]. However, the intrinsic thermal conductivity of TC4 remains relatively low($\sim 7 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$), which can lead to localized overheating and insufficient thermal shock resistance under high heat flux conditions, thereby restricting the reliability of TC4-based components in extreme environments [10], [11].

To overcome these limitations, a promising approach involves joining TC4 with silicon carbide(SiC) ceramic, which exhibits high thermal conductivity($\geq 120 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$) and a low coefficient of thermal expansion($\sim 4.5 \times 10^{-6} \text{ K}^{-1}$) [12], [13], [14], [15], [16]. This integration aims to create composite structures merging metal toughness with ceramic thermal performance. However, a significant challenge is the thermal expansion mismatch between TC4($\text{CTE} \sim 9.5 \times 10^{-6} \text{ K}^{-1}$) and SiC, which is more than double that of SiC [17], [18], [19]. This difference induces high residual thermal stress during joining, readily initiating interfacial cracks or joint fracture [20], [21], [22].

To mitigate joining reliability issues caused by thermal expansion mismatch, material modification via incorporation of high-thermal-conductivity particles (e.g., SiC) into TC4 has been explored to tailor its thermophysical properties [23], [24], [25], [26]. Conventionally, powder metallurgy (PM) processes can fabricate SiC-reinforced titanium matrix composites [27]. Theoretically, uniformly distributed SiC particles can enhance thermal conductivity while restraining matrix expansion. However, PM often results in inhomogeneous reinforcement distribution, weak interfacial bonding, and particle agglomeration, which can introduce stress concentration and adversely affect joint performance [28], [29], [30].

Additive manufacturing (AM) techniques such as selective laser melting(SLM) [31], [32] and laser cladding deposition (LCD) [33] offer new routes for fabricating composites with relatively uniform dispersion of reinforcements [34], [35]. Nevertheless, when applied to titanium matrix composites reinforced with high-melting-point ceramic particles, laser-based AM still faces

challenges like poor ceramic wettability in the melt pool, formation of brittle interfacial phases, and high residual stresses from steep thermal gradients [36], [37].

Electron beam melting (EBM) technology presents a novel approach to address these challenges [38]. Conducted in a high-vacuum environment with an electron beam heat source, the EBM process occurs at elevated temperatures, promoting slower cooling of the melt pool. This gradual solidification is beneficial for alleviating residual stresses and enhancing interfacial integrity between ceramic particles and the metal matrix [39], [40]. Furthermore, EBM enables controlled distribution of SiC within the TC4 matrix, allowing precise regulation of thermal conductivity and thermal expansion behavior [41], [42], [43]. Data from the present work show that SiC incorporation reduces the CTE difference between TC4 and SiC from $5.0 \times 10^{-6} \cdot \text{K}^{-1}$ to $2.3 \times 10^{-6} \cdot \text{K}^{-1}$, significantly mitigating interfacial stress. Leveraging the intrinsic high thermal conductivity of SiC and the formed three-dimensional interconnected structure, the composite achieves enhanced heat transfer efficiency while maintaining satisfactory mechanical properties, specifically manifested as an increase in the hardness of the matrix. Despite these advantages, systematic investigation remains lacking regarding the brazing behavior between EBM-fabricated 3D printed SiC-reinforced TC4 composites (3DP-SiC/TC4) and SiC ceramics. Particularly, the influence mechanisms of microstructure on joint performance are not yet well understood, limiting practical applications.

In this study, the brazing behavior of 3D printed materials was systematically investigated for the first time to address this research gap. A comprehensive comparison was conducted between the thermophysical properties of 3DP-SiC/TC4 and original TC4. Active AgCuTi braze was employed to join 3DP-SiC/TC4 with SiC ceramics, with systematic analysis performed on the influence of SiC reinforcements on braze wettability, interfacial reactions, and microstructural evolution. The regulatory mechanisms of brazing temperature (820°C – 880°C) on joint performance were thoroughly examined. Microstructural characterization and elemental distribution analysis were carried out using scanning electron microscopy (SEM), electron backscatter diffraction (EBSD), and transmission electron microscopy (TEM). Mechanical properties were evaluated through room-temperature shear testing. Thermal conductivity from room temperature to 600°C was measured by laser flash analysis, while transient temperature fields were captured through infrared thermography to comprehensively assess thermal performance. These findings provide novel solutions for joining 3D printed metal/ceramic dissimilar materials and are expected to facilitate practical applications in critical fields such as thermal management systems.

2. Experimental section

The 3DP-SiC/TC4 material was fabricated through electron beam melting technology, containing SiC particles ranging from 10 to 50 μm at a volume fraction of 15%, prepared in the laboratory. A 15 vol% fraction was identified as the optimal window, ensuring uniform SiC distribution for effective property enhancement while avoiding agglomeration and defects associated with higher fractions (>20%) in EBM processing. The original TC4 material was obtained from Shanghai McLean Biochemical Company with 99.8% purity and 4.43 g/cm³ density, having a chemical composition of Al (5.5–6.8 wt.%), V (3.5–4.5 wt.%), and Ti (balance). The SiC ceramic was prepared by pressureless sintering with 98.5% purity and 3.12 g/cm³ density, supplied by Hangzhou Xinfeida Technology Company. For mechanical testing and microstructural observation, 3DP-SiC/TC4 and original TC4 were machined into 15 mm $\times$ 10 mm $\times$ 5 mm specimens, while SiC ceramic was processed into 5 mm $\times$ 5 mm $\times$ 5 mm samples. Additional samples with dimensions of 10 mm $\times$ 10 mm $\times$ 1 mm were prepared from all three materials for thermal conductivity test. Prior to joining, all surfaces to be brazed were polished using 400# SiC sandpaper, ultrasonically cleaned in anhydrous ethanol for 20 min and dried. The AgCuTi braze with composition Ag-28Cu-4.5Ti (wt.%) and thickness of 100 μm was employed. The key reason for selecting AgCuTi is the active element Ti, which enables bonding to SiC by forming a Ti-Si-C layer while maintaining inherent compatibility with the TC4 matrix, complemented by the good wettability of the braze. The joint assemblies were prepared in the following order which is SiC/AgCuTi/3DP-SiC/TC4 or SiC/AgCuTi/original TC4. The orientation of the composite was not specified during brazing. Brazing was performed in a vacuum brazing furnace (Normanbell VF1300-111) under conditions of vacuum level $\leq 5 \times 10^{-3}$ Pa, applied pressure of approximately 3 kPa, heating temperature ranging from 820 $^{\circ}\text{C}$ to 880 $^{\circ}\text{C}$, and holding time of 10 min. Following brazing, samples were sectioned perpendicular to the brazing seam for subsequent grinding and polishing procedures.

Phase analysis was conducted using X-ray diffraction (XRD, Bruker D8-Advanced). Hardness measurements were performed with a nanoindentation tester (Agilent G200). The coefficient of thermal expansion was determined through thermal expansion analysis (402 Expedis Supreme). Dynamic wetting behavior was captured using a self-built wetting platform with data acquisition system. Microstructural characterization of 3DP-SiC/TC4 and joints, including distribution and morphology of reaction products, was examined by scanning electron microscopy (SEM, ZEISS Merlin Compact) equipped with energy dispersive spectroscopy (EDS). Elemental mapping across joint interfaces was performed using electron probe microanalysis (EPMA, JXA-8230). Reaction products were identified through transmission electron microscopy (TEM, FEI Talos F200S). Thermal properties including thermal conductivity, specific heat capacity, and thermal diffusivity were measured by laser flash analysis (LFA 457 NanoFlash). Thermal performance characterization is focused on the direction perpendicular to the joint, as this represents the dominant heat flow path for such brazed components during service. Shear testing was carried out using electronic universal

testing equipment. The shear strength was calculated using its standard formula,

$$\tau = P / (w \times l) \quad (1)$$

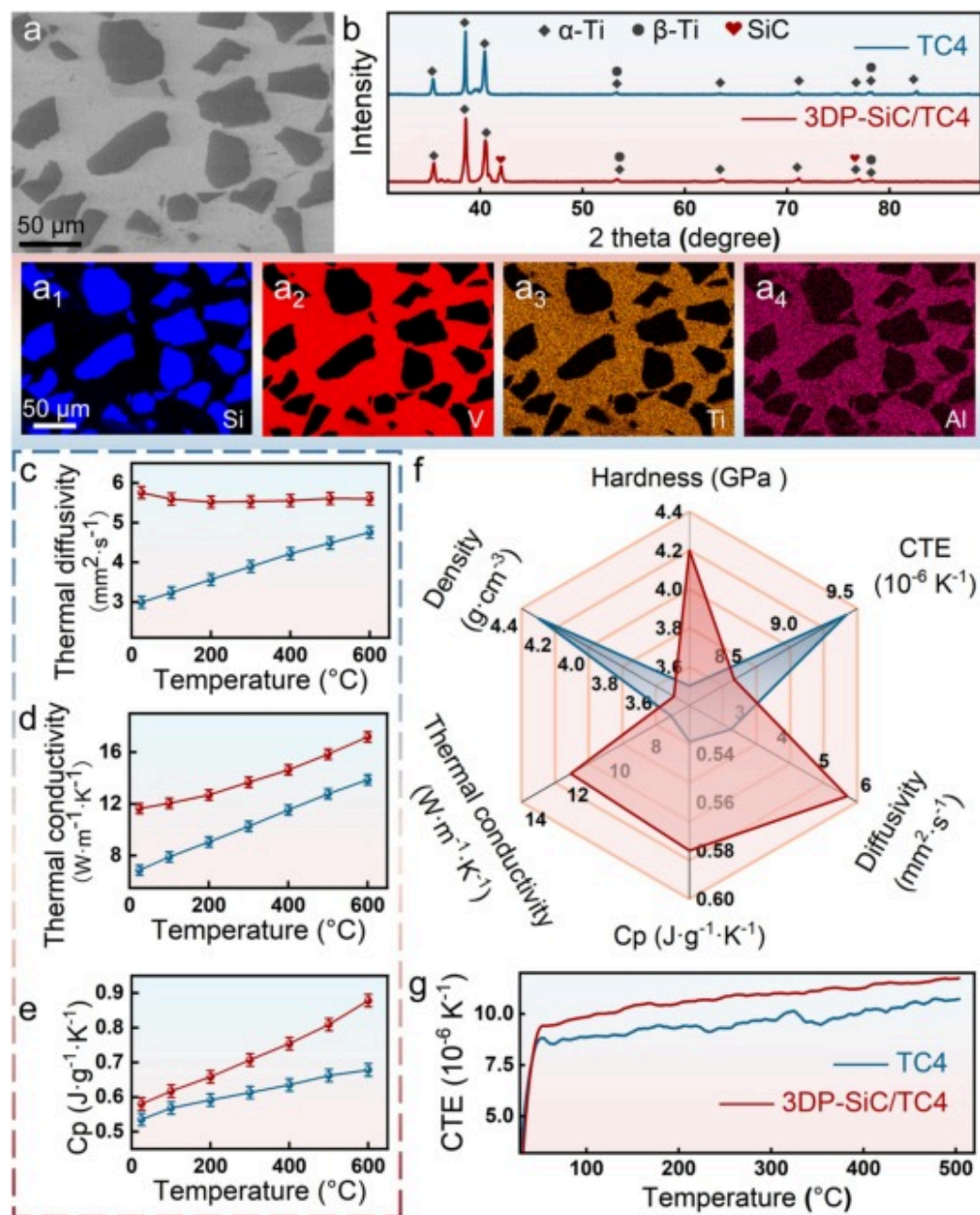
where P represents the failure load, w is the specimen width, and l is the overlap length. The loading rate was strictly set at 1.3 mm/min. For statistical reliability, five identical samples were tested for both thermal conductivity and shear strength at each brazing temperature.

3. Results and discussion

3.1. Microstructure, thermophysical properties and wetting behavior of 3DP-SiC/TC4

A comprehensive characterization of the microstructural and physical properties of original TC4 and 3DP-SiC/TC4 is presented in Fig. 1. The micrograph in Fig. 1(a) clearly demonstrates that SiC particles are uniformly distributed within the TC4 matrix. Elemental mapping analysis through EDS (Figs. 1(a₁)-1(a₄)) reveals distinct distribution patterns for elements Si, V, Ti and Al. Specifically, Si is exclusively confined within the SiC particles without noticeable interfacial diffusion, while Ti shows no detectable enrichment at the SiC interface. It should be noted that previous studies [19], [22] have demonstrated that excessive reactions between SiC and Ti would be detrimental to interfacial integrity. Therefore, suppressing interfacial reactions is essential for achieving effective bonding between these materials. This microstructural characteristic is largely attributable to the processing conditions of the EBM technology. The EBM process is conducted under high vacuum ($<5 \times 10^{-3}$ Pa) with a high preheating strategy of approximately 680°C, creating a milder thermal profile compared to laser-based additive manufacturing. This results in an extended melt pool duration at relatively lower peak temperatures, effectively suppressing excessive Ti-SiC interfacial reactions. The XRD patterns shown in Fig. 1(b) indicate that both materials are composed of α -Ti and β -Ti phases, with only minor SiC diffraction peaks being detected in 3DP-SiC/TC4. Comparative analysis of thermophysical properties in Figs. 1(c)-1(e) demonstrates that 3DP-SiC/TC4 shows superior thermophysical performance compared to original TC4 in all these aspects. A quantitative comparison of key parameters including density, hardness, thermal conductivity, specific heat capacity, coefficient of thermal expansion and thermal diffusivity is presented in the six-dimensional radar chart (Fig. 1(f)). Table 1 The uniform distribution of SiC particles within the TC4 matrix enhanced the thermal conductivity of 3DP-SiC/TC4 to 11.647 W·m⁻¹·K⁻¹. This enhancement stems from SiC's inherent high thermal conductivity (≥ 120 W·m⁻¹·K⁻¹) forming a 3D interconnected network that facilitates efficient phonon transport, while maintaining the fundamental α/β phase composition

(Fig. 1(b)). Furthermore, as illustrated in Fig. 1(g), The incorporation of SiC reinforcements leads to a reduction in the coefficient of thermal expansion for 3DP-SiC/TC4, which is particularly beneficial for subsequent brazing applications.



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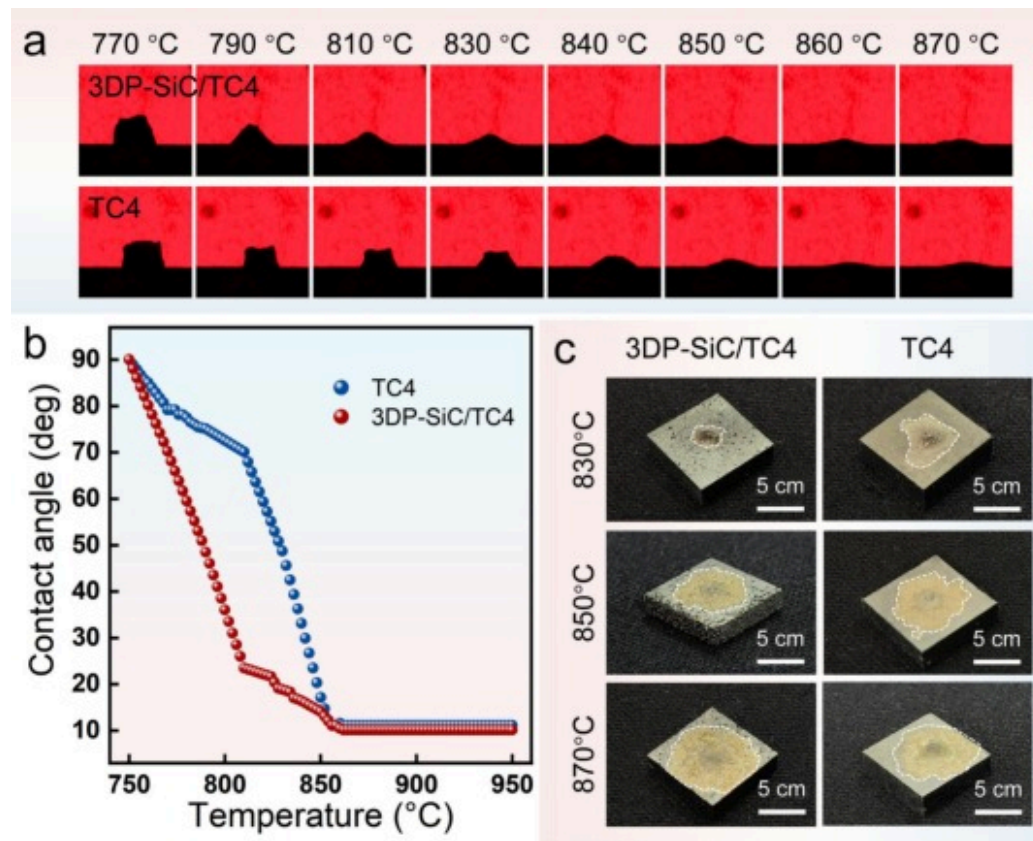
Fig. 1. Comparative characterization of microstructure and thermophysical properties between 3DP-SiC/TC4 and original TC4. (a) SEM micrograph of 3DP-SiC/TC4, (a₁)-(a₄) EDS elemental mapping, (b) XRD patterns, (c) Thermal diffusivity, (d) Thermal conductivity, (e) Specific heat capacity, (f) Radar chart comparing key parameters containing density, nanoindentation hardness, thermal conductivity, specific heat capacity, thermal expansion coefficient, and thermal diffusion coefficient, and (g) Coefficient of thermal expansion.

Table 1. Physical properties of 3DP-SiC/TC4 and TC4 in Fig. 1(f)

	3DP-SiC/TC4	TC4
Nanoindentation hardness (GPa)	4.2	3.5
CTE (10⁻⁶·K⁻¹)	8.4	9.4
Diffusivity (mm²·s⁻¹)	5.755	2.991
Cp (J·g⁻¹·K⁻¹)	0.580	0.535
Thermal conductivity (w·m⁻¹·k⁻¹)	11.647	6.873
Density (g·cm⁻³)	3.492	4.299

The wetting behavior of the AgCuTi braze on 3DP-SiC/TC4 and original TC4 substrates was investigated in Fig. 2. The active Ti in the AgCuTi braze ensured effective wetting on the prepared TC4 surfaces, overcoming any residual oxide films and confirming the reliability of the wettability tests. Fig. 2(a) illustrates the dynamic spreading process of the braze on both substrates heated from 750 °C to 870 °C at a rate of 5 °C/min. High-speed imaging revealed that rapid spreading commenced at 750 °C on the 3DP-SiC/TC4, indicating a faster response compared to the original TC4. Dynamic equilibrium was achieved at 800 °C for 3DP-SiC/TC4 with smaller contact angles, while original TC4 required heating to 830 °C to reach equilibrium. The superior wettability of 3DP-SiC/TC4 may originates from two factors: SiC reinforcement particles that reduce solid-liquid interfacial energy (γ_{SL}) to improve thermodynamic wettability, and the microporous structure created during EBM processing that provides three-dimensional capillary channels for the liquid braze spreading. This phenomenon was quantitatively characterized by contact angle-temperature curves in Fig. 2(b). Although contact angles decreased with increasing temperature for both substrates, the 3DP-

SiC/TC4 consistently exhibited lower values. Fig. 2(c) presents macroscopic morphologies at three characteristic temperatures (830 °C, 850 °C, and 870 °C). At 830 °C, on the dense TC4 substrate, the molten braze is confined to surface spreading, leading to a wide, shallow droplet with a large projected diameter. In contrast, on the porous 3DP-SiC/TC4, the braze undergoes significant capillary infiltration into the internal voids. This dominant vertical penetration consumes the braze, resulting in a surface droplet with a smaller projected area despite a much greater total wetted volume and interface. With increasing temperature, the braze on 3DP-SiC/TC4 demonstrated smaller contact angles and larger spreading areas.



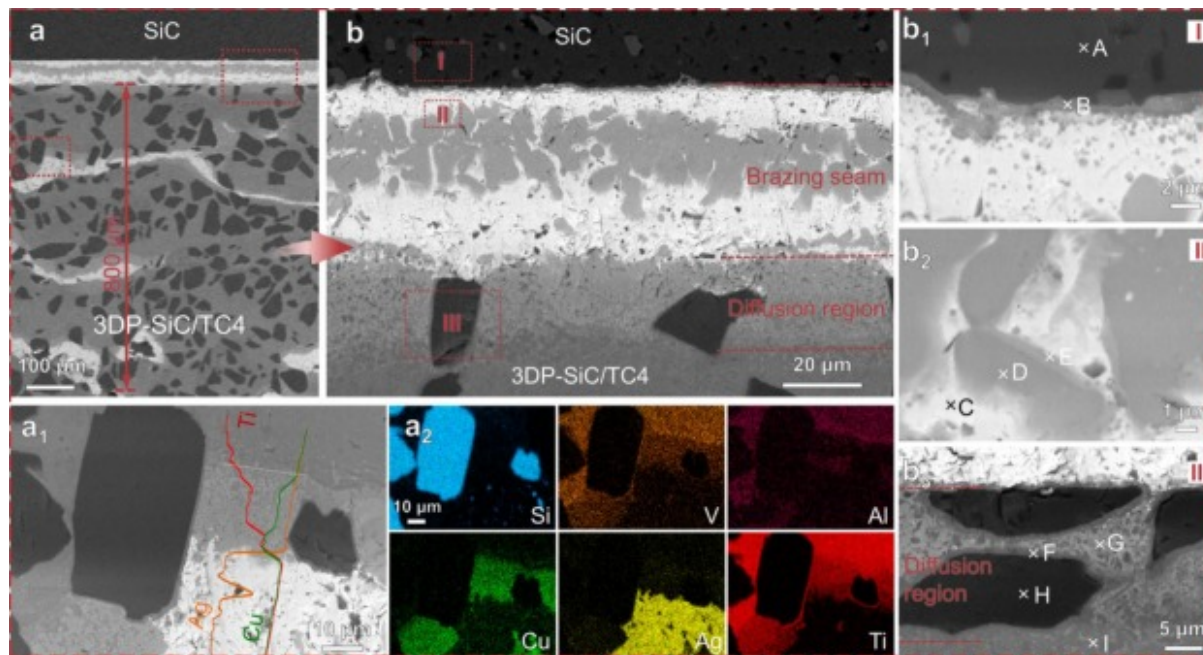
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Fig. 2. Wetting behavior of AgCuTi droplets on 3DP-SiC/TC4 and TC4 substrates. (a) Dynamic spreading process at 750-870 °C, (b) Temperature-dependent contact angle evolution, and (c) Macroscopic morphologies and cross-sectional profiles at characteristic temperatures (830 °C, 850 °C, and 870 °C).

3.2. Microstructural characteristics of 3DP-SiC/TC4/SiC brazed joints

Building upon the distinctive material characteristics of 3DP-SiC/TC4, an in-depth investigation has been conducted on SiC/3DP-SiC/TC4 brazed joints. The microstructural characteristics of SiC/3DP-SiC/TC4 joints brazed with AgCuTi braze at 840 °C for 10 min are presented in Fig. 3. Fig. 3(a) shows the overall joint morphology, where elements Ag and Cu from the braze penetrate up to 800 μm into 3DP-SiC/TC4 due to its porous nature. Statistical analysis of the Ag(s,s) infiltration depth across multiple samples yields an average of $820 \pm 100 \mu\text{m}$. The magnified view near SiC reinforcements in Fig. 3(a₁) and elemental mapping in Fig. 3(a₂) confirm the diffusion behavior of Ag and Cu, along with reactions between Cu and Ti elements. The porous structure of 3DP-SiC/TC4 facilitates deep penetration of these elements.



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Fig. 3. Microstructure characterization of SiC/3DP-SiC/TC4 joint brazed at 850 °C for 10 min. (a) Overall morphology showing Ag/Cu penetration ($\sim 800 \mu\text{m}$) in porous 3DP-SiC/TC4, (a₁) Enlarged view of SiC reinforcement phase interface, (a₂) EDS elemental maps confirming demonstrating Ag/Cu penetration into 3DP-SiC/TC4, (b) Magnified joint area with continuous brazing seam and straight interfaces, (b₁) (b₂) and (b₃) Enlarged view of Zone I, II and III in Fig. 3(b).

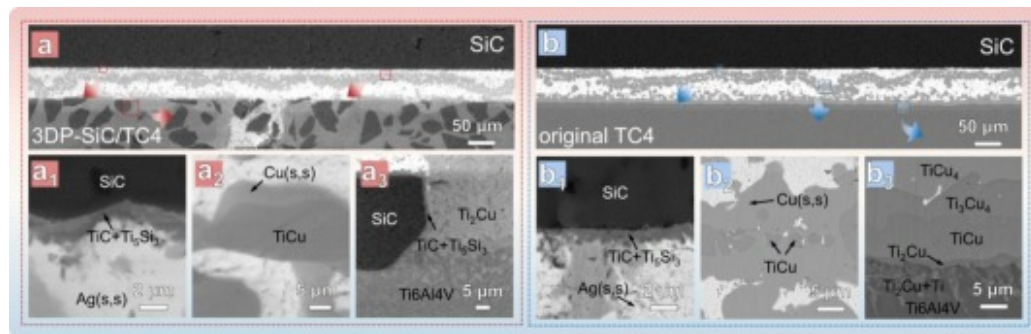
A detailed examination of the joint region in Fig. 3(b) reveals continuous and dense brazing seams with defect-free interfaces on both SiC and 3DP-SiC/TC4 sides. In region I of the brazing seam (Fig. 3(b₁)), a Ti-Si-C reaction layer approximately 1 μm thick forms at the SiC interface, potentially containing brittle phases such as TiC, and Ti₅Si₃. Phase A is identified as intact SiC matrix and phase B as TiC+Ti₅Si₃ by EDS analysis shown in Table. 2. The formation of this reaction layer results from vigorous interfacial reactions between Ti and SiC, which directly influence joint mechanical properties. However, the continuous and uniform morphology suggests effective control of brittle phase growth during brazing with 3DP-SiC/TC4. Three typical phases are observed in region II of the brazing seam (Fig. 3(b₂)): silver-white phase C (Ag(s,s)), gray-white phase D (Cu Ag(s,s)), and light-gray phase E (Cu-Ti intermetallic compounds). The Cu-Ti intermetallics dominate, with Cu(s,s) surrounding them. Region III (Fig. 3(b₃)) displays a wide Cu-Ti diffusion zone on the 3DP-SiC/TC4 side, where dark-gray phase F shows mixed characteristics of Ti₅Si₃ and TiC as reaction products at SiC reinforcement interfaces. Within this diffusion zone, gray-white phase G (Ti₂Cu) and gray-black phase I (Ti-6Al-4V matrix) are clearly identified. Notably, braze components exhibit distinct gradient diffusion patterns in this region (Fig. 3(b)). The gradient distribution of interfacial reaction products, particularly the Cu-Ti diffusion zone on the 3DP-SiC/TC4 side, demonstrates that the deep diffusion effects enabled by porous structures provide new perspectives for brazing 3D printed materials.

Table 2. The EDS analysis results of the various phases in the joint are shown in Fig. 3(at. %).

Points	C	Si	Ti	Al	V	Ag	Cu	PossiblePhase
A	56.5	43.5	-	-	-	-	-	SiC
B	18.5	30.2	49.8	-	-	1.0	0.5	Ti ₅ Si ₃ +TiC
C	-	3.6	-	-	-	86.7	9.8	Ag(s,s)
D	-	-	20.6	-	-	2.3	77.1	Cu(s,s)
E	-	-	49.5	-	-	2.0	48.0	TiCu
F	44.2	28.3	27.5	-	-	-	-	Ti ₅ Si ₃ +TiC
G	-	-	62.1	4.8	1.2	2.4	29.5	Ti ₂ Cu
H	54.6	45.4	-	-	-	-	-	SiC

Points	C	Si	Ti	Al	V	Ag	Cu	PossiblePhase
I	-	-	87.2	10.5	2.3	-	-	Ti6Al4V

Fig. 4 presents a microstructural comparison between SiC/AgCuTi/3DP-SiC/TC4 and SiC/AgCuTi/original TC4 joints brazed at 850 °C for 10 min. SEM images in Fig. 4(a) (3DP-SiC/TC4) and Fig. 4(b) (original TC4) reveal that while both joints exhibit continuous and dense brazing seams, significant differences exist in interfacial reactions, phase distribution, and elemental diffusion. First, at the SiC interface, the Ti-Si-C reaction layer formed in the 3DP-SiC/TC4 joint (Fig. 4(a₁)) appears more compact and continuous with slightly greater thickness compared to the original TC4 joint (Fig. 4(b₁)). Second, in the Ag-rich brazing seam region, Cu exists primarily as Cu-Ti intermetallic compounds in the 3DP-SiC/TC4 joint (Fig. 4(a₂)), whereas Cu(s,s) dominates in the original TC4 joint (Fig. 4(b₂)), with only minor Cu-Ti intermetallics forming at the center. This phase distribution results from strong Ti diffusion from the substrate into the braze, reacting with Cu to form stable Ti-Cu compounds. Higher Ti-Cu compound content indicates greater Ti consumption through reactions with Cu, thereby effectively regulating Ti-Si-C reactions at the SiC interface. Third, the TC4-side diffusion zone in the 3DP-SiC/TC4 joint (Fig. 4(a₃)) demonstrates superior Cu diffusion characteristics, exhibiting a gradient transition over approximately 20 μm, progressing from a continuous Ti₂Cu layer to a dispersed (α+β)Ti particle region and finally to the (α+β)Ti matrix. In contrast, the original TC4 joint (Fig. 4(b₃)) shows only a thin 4 μm Ti₂Cu layer without significant gradient transition. This difference originates from the unique porous structure of 3DP-SiC/TC4, where interlayer gaps and SiC reinforcement/matrix interfaces created during EBM processing provide fast diffusion channels for Cu. While SiC particles locally alter reaction morphology by promoting TiC/Ti₅Si₃ formation (Fig. 3(a₁), Fig. 4(a₃)), the overall interfacial layer remains continuous (Fig. 3(b₁)), indicating that their influence on the integrity of the joint is secondary to the uniform reaction in the matrix. The preliminary porosity in EBM-fabricated 3DP-SiC/TC4 predominantly governs the infiltration depth of Ag and Cu. The porous network acts as capillary channels, enabling deep penetration of the Ag(s,s) phase up to 800 μm, as evidenced in Fig. 3(a). In contrast, Cu diffusion is primarily driven by interfacial reaction with the TC4 matrix, forming a Ti-Cu intermetallic zone (Fig. 4(a₃)), which is less dependent on the bulk porosity.

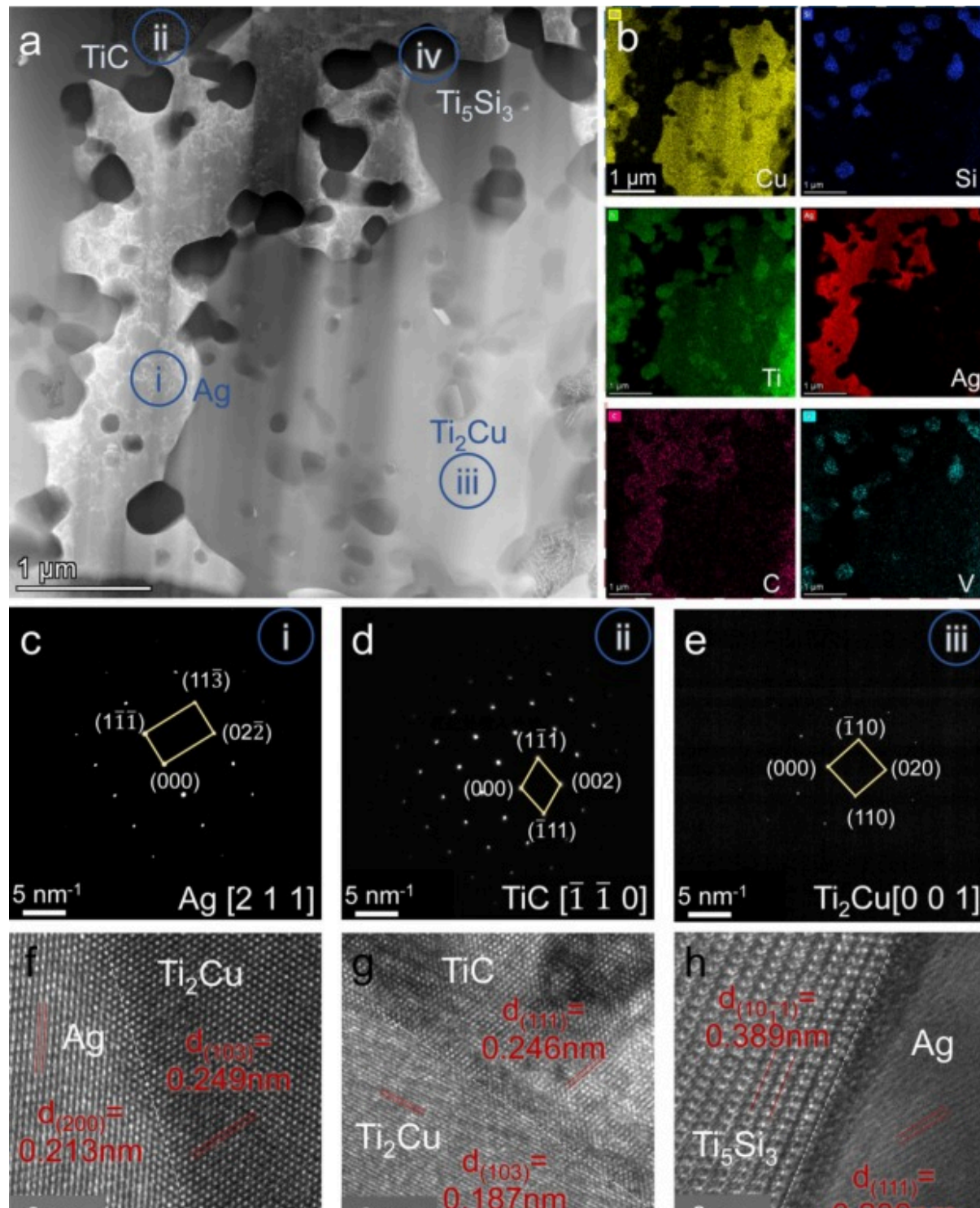


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Fig. 4. Microstructural comparison between SiC/AgCuTi/3DP-SiC/TC4 and SiC/AgCuTi/original TC4 joints brazed at 850 °C for 10 min. (a) The SEM image of the SiC/AgCuTi/3DP-SiC/TC4 joint brazed at 850 °C for 10 min. (b) The SEM image of the SiC/AgCuTi/original TC4 joint brazed at 850 °C for 10 min. (a₁), (a₂) and (a₃) Magnification of the regions in (a). (b₁), (b₂) and (b₃) Magnification of the regions in (b).

The interfacial microstructure near SiC reinforcements in 3DP-SiC/TC4/SiC joints brazed at 850 °C for 10 min was characterized by TEM, as shown in Fig. 5. The HAADF-STEM image in Fig. 5(a) clearly reveals multiphase structural features at the interface, where four characteristic regions can be identified through image analysis. Region (i) exhibits bright-contrast Ag(s,s) phase, confirming Ag infiltration into 3DP-SiC/TC4. Region (ii) contains minor TiC phase formed by interfacial reaction between SiC and Ti. Region (iv) displays Ti₅Si₃ phase resulting from SiC-Ti reaction. Region (iii) shows bright-contrast Cu-Ti compound phase, identified as Ti₂Cu based on EDS results. STEM-EDS elemental mapping in Fig. 5(b) further verifies the chemical composition characteristics of these phases. Significant Cu enrichment in region (iii) confirms Ti₂Cu intermetallic formation. Si is predominantly distributed in Ti₅Si₃ phase, while trace V presence originates from diffusion of TC4 matrix (Ti-6Al-4V). It should be noted that Ti participates in both Ti₅Si₃ and TiC phase formation, with C signals mainly from TiC, indicating Ti-C compound generation at SiC reinforcement interface that was not observed in previous studies.





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Fig. 5. TEM analysis of the interface near the SiC reinforcement phase in 3DP-SiC/TC4 brazed at 850 °C for 10 min. (a) HAADF image of the 3DP-SiC/TC4 interface, (b) STEM EDS mapping of region in (a), (c) SAED patterns of area i in (a), (d) SAED patterns of area ii in (a), (e) SAED patterns of area iii in (a), (f) HRTEM image of Ag/Ti₂Cu interface, (g) HRTEM image of Ti₂Cu/TiC interface, and (h) HRTEM image of Ag/Ti₅Si₃ interface.

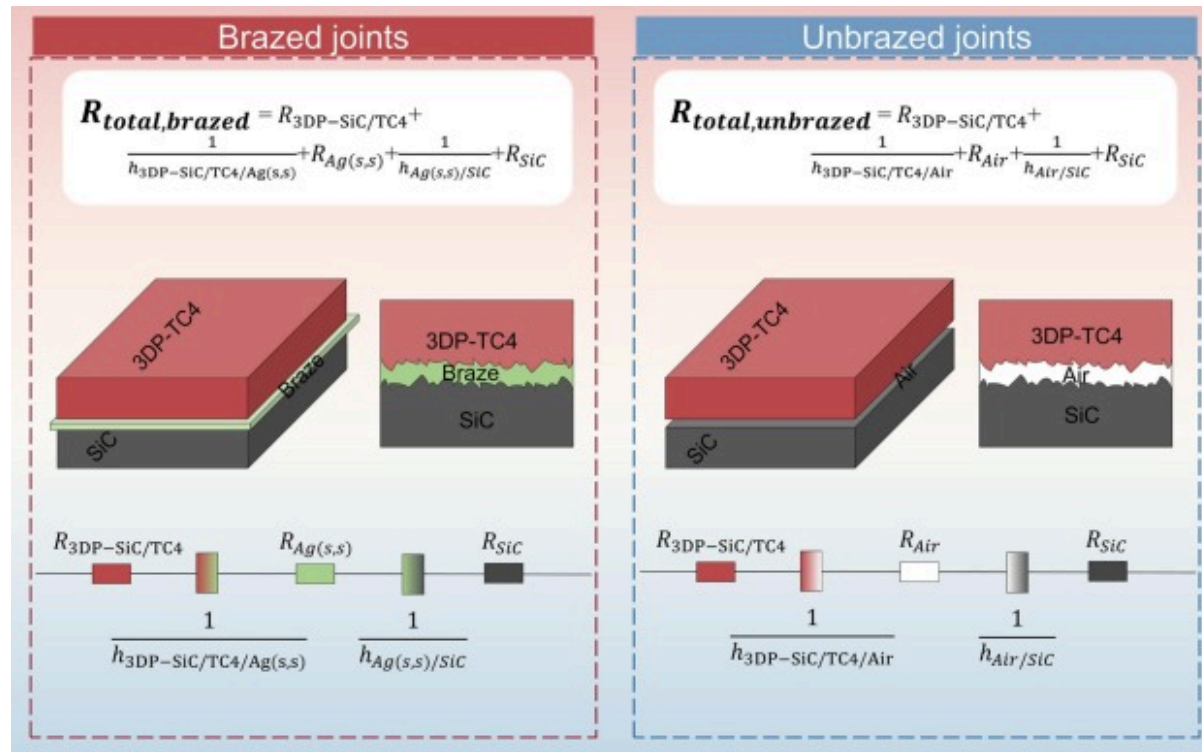
The crystal structures of Ag, TiC and Ti₂Cu were precisely determined through selected area electron diffraction analysis (Figs. 5(c)-5(e)). In Fig. 5(c), the characteristic diffraction pattern of face-centered cubic Ag along the [211] zone axis is presented, with the lattice parameter measured to be 0.408 nm. The diffraction pattern of TiC along the $[\bar{1}\bar{1}0]$ zone axis in Fig. 5(d) confirms the existence of cubic TiC phase. Fig. 5(e) displays the diffraction pattern of tetragonal Ti₂Cu phase along the [001] zone axis, verifying that the Ti-Cu intermetallic compound is indeed Ti₂Cu.

Atomic-scale interfacial characteristics were revealed by high-resolution TEM analysis (Figs. 5(f)-5(h)). The Ag/Ti₂Cu interface structure is displayed in Fig. 5(f), where the measured interplanar spacing of Ti₂Cu (103) planes (0.249 nm) forms a coherent interface with adjacent Ag (200) planes (0.213 nm spacing). Fig. 5(g) presents the Ti₂Cu/TiC interface structure, showing measured TiC(111) plane spacing of 0.246 nm compared to the theoretical value of 0.250 nm. The Ag/Ti₅Si₃ interface in Fig. 5(h) exhibits measured Ti₅Si₃ (101) plane spacing of 0.213 nm versus the theoretical 0.215 nm, indicating minor deviation. These interplanar spacing differences suggest slight lattice distortion at interfaces that may serve as potential crack initiation sites. The small deviations demonstrate limited lattice mismatch and strong interfacial joining at the 850 °C brazing temperature.

The aforementioned research systematically reveals the microstructural formation mechanisms of 3DP-SiC/TC4/SiC brazed joints and their regulatory effects on interfacial performance. To achieve comprehensive performance optimization in heterogeneous material joining, further elucidation is required regarding how microstructural characteristics influence thermal transport properties.

3.3. Thermal conductivity of the joint

As is well known, heat transfer in 3DP-SiC/TC4 is mainly accomplished by free electrons. But in SiC, it is mainly achieved by phonons. The use of brazing alloys makes the microstructure of the interface more complex. At the interface, the scattering effect is enhanced during the transfer of heat carriers, undoubtedly resulting in a significant interface thermal resistance. Therefore, it is necessary to study the heat transfer mechanism of 3DP-SiC/TC4/SiC brazed joints (as shown in Fig. 6).



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Fig. 6. Equivalent thermal resistance of the 3DP-SiC/TC4 brazed joint and the unbrazed joint.

The analysis of joint thermal resistance distribution employs a layered thermal resistance model, where the total thermal resistance is equal to the sum of the thermal resistances of each material and the thermal resistances at each interface. The

formula is as follows [44], [45]:

$$R_{\text{total}} = \sum \left(\frac{\delta}{K} \right) + \sum \left(\frac{1}{h_{ij}} \right) \quad (2)$$

For a homogeneous material layer, the physical relationship between the bulk thermal resistance and its thermal conductivity (K) and thickness (δ) of the material can be expressed as:

$$R_{\text{bulk}} = \frac{\delta}{K} \quad (3)$$

For the two-phase interface, the physical relationship between thermal resistance and the interfacial heat transfer coefficient h can be expressed as:

$$R_{\text{interface}} = \frac{1}{h} \quad (4)$$

Therefore, for the 3DP-SiC/TC4/SiC brazed joint, the total thermal resistance $R_{\text{total, brazed}}$ can be simplified as:

$$R_{\text{total, brazed}} = R_{\text{3DP-SiC/TC4}} + R_{\text{Ag(s,s)}} + R_{\text{SiC}} + \frac{1}{h_{\text{3DP-SiC/TC4/Ag(s,s)}}} + \frac{1}{h_{\text{Ag(s,s)/SiC}}} \quad (5)$$

Among them, the brazing seam composition was simplified to 3DP-SiC/TC4/Ag(s,s) and Ag(s,s)/SiC, account for two primary interfaces. Similarly, $R_{\text{total, unbrazed}}$, the total thermal resistance for unbrazed joints can be expressed as:

$$R_{\text{total, unbrazed}} = R_{\text{3DP-SiC/TC4}} + R_{\text{Air}} + R_{\text{SiC}} + \frac{1}{h_{\text{3DP-SiC/TC4/Air}}} + \frac{1}{h_{\text{Air/SiC}}} \quad (6)$$

The interfacial heat transfer coefficient is calculated through a phonon transport model:

$$h_{i/j} = \frac{\rho_i c_i v_i^3}{4(v_i^2 + v_j^2)} \quad (7)$$

In the Eqn 7, ρ_i , c_i , v_i represent material density, specific heat capacity, and sound velocity, respectively. The sound velocity v is calculated through the definition formula of thermal diffusivity:

$$v = \sqrt{\alpha/\rho} \quad (8)$$

In the Eqn 8, α represents the thermal diffusivity. The parameters required for calculating the heat transfer coefficient $h_{i/j}$ of each

two-phase interface and the thermal resistance of the material are shown in Table 3, and the results obtained based on these parameters are also presented in Table 3.

Table 3. Material parameter table for each layer

Layer	$\delta(\text{mm})$	$K(\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1})$	$\alpha(\text{mm}^2\cdot\text{s}^{-1})$	$c(\text{J}\cdot\text{g}^{-1}\cdot\text{K}^{-1})$	$\rho(\text{kg}\cdot\text{m}^{-3})$	$\nu(\text{m}\cdot\text{s}^{-1})$	$R_{\text{bulk}}(\times 10^{-6}\text{m}^2\cdot\text{K}\cdot\text{W}^{-1})$
3DP-SiC/TC4	1.0	11.647	5.755	0.580	3.492	4050	85.9
Ag(s,s)	0.04	429	100	0.235	10.49	2000	0.093
Air	0.01	0.026	19.1	1.005	1.225	343	384.6
SiC	1.0	120	65	0.70	3.21	4500	8.33

Using the parameters in Table 3, the interface heat transfer coefficients h_{ij} of each two-phase interface calculated according to the formula are shown in Table 4:

Table 4. Calculation results of interfacial heat transfer coefficient h_{ij} and $R_{\text{interface}}$

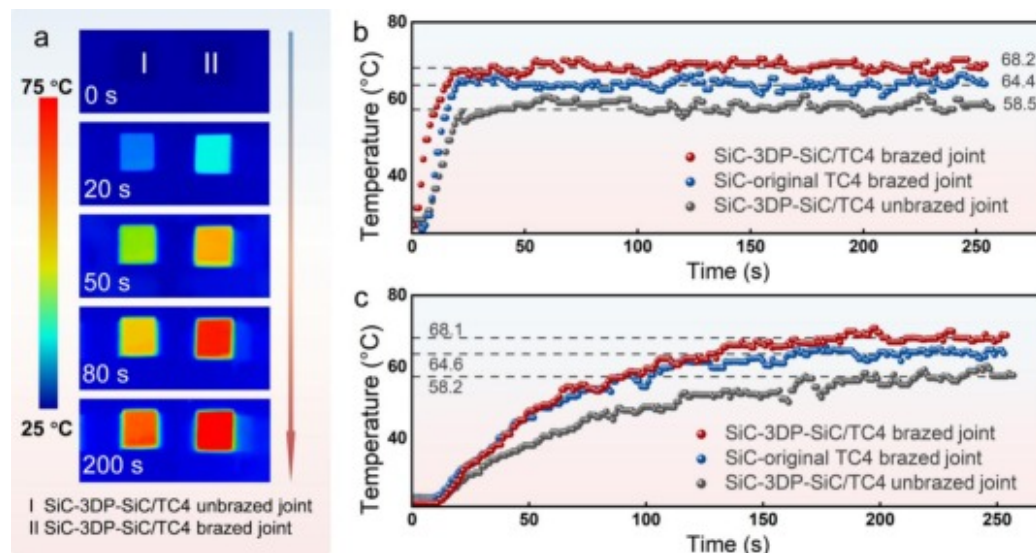
Interlayer	$h_{ij}(\text{W}\cdot\text{m}^{-2}\cdot\text{K}^{-1})$	$R_{\text{interface}}=1/h_{ij}(\times 10^{-6}\text{m}^2\cdot\text{K}\cdot\text{W}^{-1})$
3DP-SiC/TC4/Ag(s,s)	2.85×10^7	0.035
Ag(s,s)/SiC	1.02×10^7	0.098
3DP-SiC/TC4/Air	3.01×10^6	0.332
Air/SiC	1.20×10^4	83.3

The thermal resistance values were calculated by substituting the obtained data into Eqn 5 and Eqn 6 using a layered thermal resistance model. The thermal resistance analysis based on a layered model (as shown in Fig. 6) reveals significant improvements in 3DP-SiC/TC4/SiC joints through brazing. The total thermal resistance of the brazed joint was measured at $94.5 \times 10^{-6}\text{m}^2\cdot\text{K}\cdot\text{W}^{-1}$, representing an 83.2% reduction compared to the unbrazed joints ($562.5 \times 10^{-6}\text{m}^2\cdot\text{K}\cdot\text{W}^{-1}$). This substantial decrease primarily

results from the replacement of air gaps by Ag braze layers, which reduced the joint gap resistance from $384.6 \times 10^{-6} \text{ m}^2 \cdot \text{K} \cdot \text{W}^{-1}$ to merely $0.093 \times 10^{-6} \text{ m}^2 \cdot \text{K} \cdot \text{W}^{-1}$. The brazing process effectively eliminates air voids and enhances interfacial joining, thereby addressing the thermal resistance caused by air layers and Air/SiC interfaces in unjoined assemblies. Furthermore, metallic interfaces formed during brazing significantly improve phonon transport efficiency. The heat transfer coefficients at 3DP-SiC/TC4/Ag(s,s) and Ag(s,s)/SiC interfaces reached $2.85 \times 10^7 \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$ and $1.02 \times 10^7 \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$ respectively, representing improvements of 1-3 orders of magnitude compared to unbrazed interfaces. The combined interfacial thermal resistance was only $0.133 \times 10^{-6} \text{ m}^2 \cdot \text{K} \cdot \text{W}^{-1}$, demonstrating superior performance over conventional mechanical contacts.

Critically, although Ti-Cu intermetallic compound has moderate thermal conductivity, its integration within a composite interfacial structure containing highly conductive Ag(s,s) and TiC phases results in a net reduction of the total thermal resistance by 83.2%, demonstrating its non-decisive impact. Additionally, the thermal resistance distribution in brazed joints exhibits an optimal configuration. The bulk resistance of 3DP-SiC/TC4 ($85.9 \times 10^{-6} \text{ m}^2 \cdot \text{K} \cdot \text{W}^{-1}$) accounts for 90.9% of the total, while the SiC layer contributes $8.33 \times 10^{-6} \text{ m}^2 \cdot \text{K} \cdot \text{W}^{-1}$ (8.8%). Interfaces collectively represent merely 0.14% ($0.133 \times 10^{-6} \text{ m}^2 \cdot \text{K} \cdot \text{W}^{-1}$) of the total resistance. This distribution pattern confirms a bulk-material-dominated heat transfer mechanism, where joint performance is primarily determined by intrinsic thermal conductivity properties with negligible interfacial contributions.

Fig. 7 demonstrates the superior heat transfer characteristics of SiC-3DP-SiC/TC4 brazed joints through multiscale thermal analysis. The infrared thermal image in **Fig. 7(a)** reveals transient temperature fields, clearly showing thermal response differences among three joint types. The SiC-3DP-SiC/TC4 brazed joint exhibits significantly faster heating rates than the SiC-3DP-SiC/TC4 unbrazed joint under thermal excitation. This advantage originates from AgCuTi braze filling air gaps within and between base materials, eliminating air voids present in unbrazed samples and establishing three-dimensional thermal conduction networks. Steady-state heat transfer behavior at 100 °C constant heating (**Fig. 7(b)**) and non-steady-state heating from room temperature to 100 °C (**Fig. 7(c)**) were examined. During the initial transient stage (0-30 s), the SiC-3DP-SiC/TC4 brazed joint displays the steepest temperature rise slope, indicating fastest thermal response. This notable kinetic difference suggests higher interfacial heat transfer efficiency in the SiC-3DP-SiC/TC4 brazed joints, primarily attributed to braze elimination of interfacial air gap thermal resistance present in unbrazed samples. Additionally, SiC reinforcement incorporation enhances intrinsic thermal conductivity of 3DP-SiC/TC4 composites, resulting in higher post-brazing joint thermal conductivity compared to original TC4 joints. The comprehensive thermal analysis confirms that both material composition and joining quality contribute to enhanced heat transfer performance in 3DP-SiC/TC4 brazed joints compared to original TC4 brazed joints.



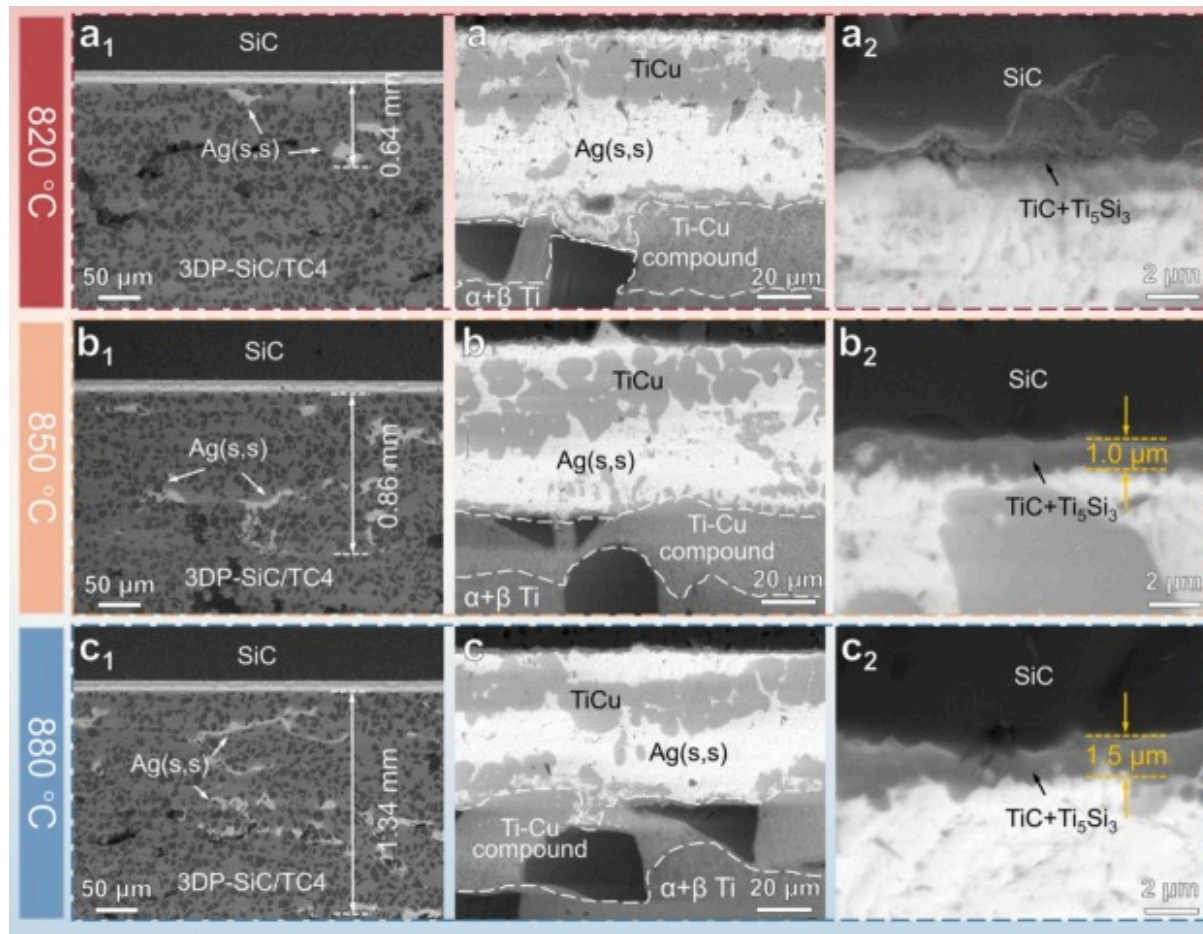
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Fig. 7. Thermal transport characterization of brazed joints. (a) Infrared thermography showing temperature field evolution of the upper surfaces of the sample, (b) Upper surface temperature of plate versus time under steady-state heat conduction at 100 °C, (c) Upper surface temperature of the sample versus time under unsteady-state heat conduction heating from RT to 100 °C.

3.4. Effect of the brazing temperature on microstructure, thermal conductivity and shear strength of the joint

The influence of brazing temperature (820 °C–880 °C) on the microstructure of SiC/AgCuTi/3DP-SiC/TC4 joints was systematically investigated in Fig. 8. A decreasing trend in brazing seam width was observed with increasing temperature. As shown in Fig. 8(a₁), the maximum brazing seam width occurred at 820 °C, while gradual reduction was observed at 850 °C (Fig. 8(b₁)) and 880 °C (Fig. 8(c₁)). Concurrently, Ag diffusion depth increased significantly with temperature, measuring approximately 0.64 mm at 820 °C and reaching 1.34 mm at 880 °C. This variation originates from reduced braze viscosity at elevated temperatures, promoting improved wetting and diffusion.



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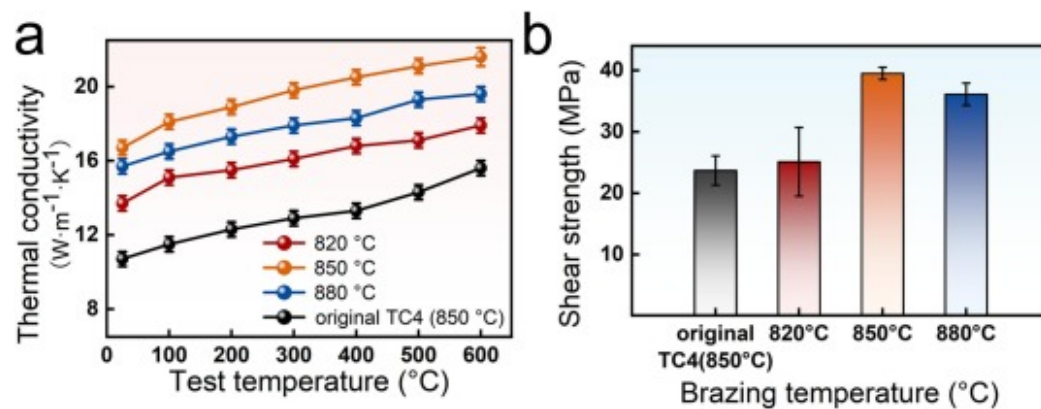
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Fig. 8. The SEM image of the SiC/AgCuTi/3DP-SiC/TC4 joint brazed for 10 min at (a) 820 °C, (b) 850 °C and (c) 880 °C. The SEM image of braze penetration in 3DP-SiC/TC4 brazed for 10 min at (a₁) 820 °C, (b₁) 850 °C and (c₁) 880 °C. (a₂), (b₂) and (c₂) Magnification of the regions near the SiC interface in (a), (b) and (c), respectively.

Temperature-dependent microstructural evolution was observed on the 3DP-SiC/TC4 side, as illustrated in Figs. 8(a)-8(c). The thickness of Ti-Cu compound layers increased with temperature due to enhanced Ti-Cu reactions and deeper Cu diffusion into 3DP-SiC/TC4 at higher temperatures. SiC interfacial reaction layers exhibited temperature-dependent characteristics shown in

Figs. 8(a₂)-8(c₂). At 820 °C, insufficient reaction led to discontinuous Si-Ti-C layers with unreacted SiC regions and microvoids at interfaces. The reaction layer became continuous with 1.0 μm thickness at 850 °C, further growing to 1.5 μm at 880 °C, indicating strengthened metallurgical joining through increased Si-Ti-C reaction layer thickness and continuity. However, excessive growth of brittle Si-Ti-C compounds at higher temperatures may introduce microcracks and reduce interfacial toughness, potentially compromising joint integrity. The accelerated interfacial reaction and formation of a thick, brittle layer at 880°C (Fig. 8(c₂)) define a practical upper-temperature limit, beyond which excessive degradation would jeopardize the long-term stability of the joint for high-temperature service. An optimal reaction layer thickness of ~1.0 μm (Fig. 3(b₁)), achieved at 850°C by controlled Ti activity, is critical for balancing interfacial bonding and avoiding brittleness. These findings demonstrate the critical balance required in temperature selection to optimize interfacial characteristics while avoiding detrimental brittle phase accumulation.

As detailed in Fig. 9(a), brazing temperature significantly influenced the thermal conductivity of the joint. Optimal performance was achieved at 850°C, with values reaching 16.7 W·m⁻¹·K⁻¹ at room temperature and 21.6 W·m⁻¹·K⁻¹ at 600°C, representing a 56% improvement over joints made with original TC4, indicating robust stability of the interfacial structure, including the Ti-Cu phases, under elevated temperatures. A similar trend was observed for mechanical properties. The shear strength, shown in Fig. 9(b), also peaked at 850°C with a maximum value of 39.5±1.0 MPa. The precise control of the interfacial reaction layer thickness to ~1.0 μm (Fig. 8(b₂)) prevented microcracking (Fig. 3(b₁), Fig. 4(a₁)) despite CTE mismatch, which is a key factor enabling the high shear strength of 39.5±1.0 MPa (Fig. 9(b)). The soft Ag-rich phase acted as a primary energy-dissipation mechanism through plastic deformation, and the robust Ag/Ti₂Cu interface ensured effective load transfer. This was 57% higher than joints brazed at 820°C and 9% higher than those at 880°C, and substantially exceeded the strength of original TC4 joints (23.7±2.4 MPa).

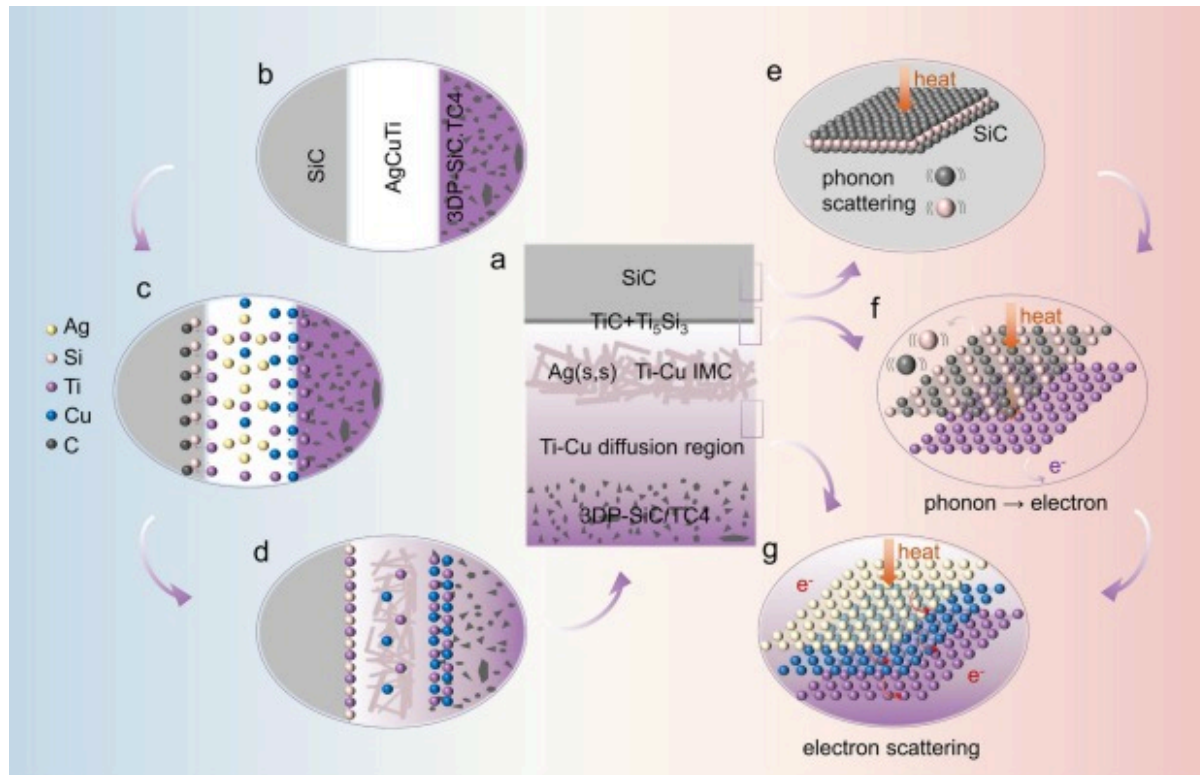


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Fig. 9. Thermal and mechanical properties of SiC/AgCuTi/3DP-SiC/TC4 brazed joints. (a) Temperature-dependent thermal conductivity of original TC4 and 3DP-SiC/TC4 joints brazed at 820 °C, 850 °C, and 880 °C, and (b) Room-temperature shear strength comparison among original TC4 joints (brazed at 850 °C for 10 min) and brazed 3DP-SiC/TC4 joints (brazed at 820 °C, 850 °C, and 880 °C for 10 min).

The microstructural evolution and thermal transport mechanisms in 3DP-SiC/TC4/SiC brazed joints were systematically investigated, as presented in Fig. 10. The schematic representation of brazed microstructure in Fig. 10(a) clearly illustrates the layered architecture comprising SiC ceramic phase, Ti-Si-C reaction layer, Ag(s,s) and Ti-Cu IMC region, Ti-Cu diffusion region, and 3DP-SiC/TC4 matrix. The initial assembly configuration is shown in Fig. 10(b). During the early heating stage (Fig. 10(c)), Ti migrates toward both the braze center and SiC interface, while Cu from the braze diffuses into the 3DP-SiC/TC4 substrate. With increasing temperature (Fig. 10(d)), Ti diffuses towards the SiC interface to participate in the formation of the layered TiC/Ti₅Si₃ reaction zone. Simultaneously, it reacts with Cu at the braze center to form Ti-Cu IMCs. The diffusion of Cu into the TC4 matrix produces graded Ti-Cu intermetallic layers, which effectively consumes Ti from the substrate and thereby regulates the extent of interfacial reactions at the SiC side.



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Fig. 10. The microstructural evolution and thermal transport mechanisms in 3DP-SiC/TC4/SiC brazed joints.

The microscopic heat transfer mechanisms within the joint are elucidated in Figs. 10(e)-10(g). As shown in Fig. 10(e), heat conduction occurs primarily through phonon transport on the SiC ceramic side. In contrast, Fig. 10(g) demonstrates that heat transfer in the 3DP-SiC/TC4 side relies mainly on free electron movement through the titanium alloy matrix. Particular attention should be given to the carrier transition mechanism in the interfacial transition zone (Fig. 10(f)). At the SiC/TiC/Ti₅Si₃/Ag(s,s) interface, thermal carriers must transition from phonon-dominated mode in SiC to electron-dominated mode in metal through electron-phonon coupling. This process is mediated by the intervening TiC and Ti₅Si₃. Heat arrives from the phonon-dominated SiC ceramic side. At the SiC/TiC interface, due to the strong covalent bonding nature shared by both materials and the partial overlap in their phonon spectra, phonons can be relatively easily injected into TiC. Subsequently, within the TiC/Ti₅Si₃/Ag(s,s)

region, a gradual transition from ceramic-like to metal-like character exists. The free electrons in Ag(s,s) interact with the lattice vibration phonons of TiC and Ti_5Si_3 , a process known as electron-phonon coupling. This interaction converts phonon energy into electron kinetic energy, thereby accomplishing the relay of heat transfer. The introduction of a TiC/ Ti_5Si_3 transition layer reduces lattice mismatch and enhances conversion efficiency. Optimal control of the Ti-Si-C reaction layer thickness ($<1\text{ }\mu\text{m}$) minimizes phonon scattering while improving overall joint thermal conduction efficiency.

Comprehensive analysis demonstrates that the optimized brazing process achieves at $850\text{ }^\circ\text{C}$. First, regarding microstructure, continuous Ti-Si-C layers rather than discrete brittle phases ensure interfacial joining strength. Second, concerning heat transfer, appropriately thick Ti-Si-C layers effectively reduce phonon scattering, facilitate electron-phonon coupling transition, and maintain high interfacial thermal conductivity, while avoiding excessive phonon scattering caused by overgrown intermetallic compounds. This study not only provides process optimization guidelines for 3DP-SiC/TC4/SiC heterogeneous joining but also offers significant reference for interface design and performance control in other ceramic-3D printed metal systems.

4. Conclusions

In this study, high-quality brazed joints between 3DP-SiC/TC4 and SiC ceramics were successfully achieved using the AgCuTi braze, demonstrating excellent mechanical and thermal properties.

(1) Microstructure, physical properties, and wettability of SiC-reinforced Ti6Al4V composites (3DP-SiC/TC4) fabricated by electron beam melting (EBM) were systematically investigated. Compared with original TC4, 3DP-SiC/TC4 exhibits a lower coefficient of thermal expansion ($8.4\times 10^{-6}\cdot\text{K}^{-1}$) and a higher thermal conductivity ($11.647\text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$). Additionally, the AgCuTi braze demonstrates faster spreading velocity and smaller contact angle on the 3DP-SiC/TC4 surface.

(2) Effective joining between SiC and 3DP-SiC/TC4 was accomplished. The porous structure of 3DP-SiC/TC4 significantly enhances the penetration of the liquid braze and the elemental diffusion, with partial Ag(s,s) diffusing into the 3DP-SiC/TC4 matrix. A continuous reaction layer ($\sim 1.0\text{ }\mu\text{m}$ thickness) forms preferentially along the SiC interface, consisting of a TiC layer adjacent to the SiC and a Ti_5Si_3 layer towards the brazing seam, ensuring a strong interfacial joining. Thermodynamic calculations and experimental characterization jointly confirm the preferential occurrence of Ti-Si reactions. The central braze region consists mainly of Ag(s,s), while Cu reacts with Ti to form TiCu intermetallic compounds. A graded Ti-Cu diffusion zone ($\sim 20\text{ }\mu\text{m}$ wide) develops on the 3DP-SiC/TC4 side.

(3) Compared with original TC4 joints, 3DP-SiC/TC4 joints exhibit distinct characteristics. First, Cu in the braze center exists primarily as intermetallic compounds rather than solid solutions. Second, 3DP-SiC/TC4 joints feature a unique Ti-Cu gradient diffusion structure, with Cu penetration depth (20 μm) five times greater than in original TC4 joints (4 μm), forming a wider transition zone. The shear strength (39.5 ± 1.0 MPa) and thermal conductivity ($21.6 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$) of 3DP-SiC/TC4 joints are 57% and 56% higher than those of original TC4 joints, respectively.

(4) Enhanced thermal transport performance originates from the intrinsically higher thermal conductivity of 3DP-SiC/TC4. The Ti-Si-C transition layer facilitates carrier conversion by acting as a phonon bridge from the SiC, thereby reducing lattice mismatch and enhancing electron-phonon coupling to the metal, which effectively reduces the interfacial thermal resistance to $94.5 \times 10^{-6} \text{ m}^2 \cdot \text{K} \cdot \text{W}^{-1}$.

(5) Brazing temperature significantly influences joint mechanical and thermal properties by altering the microstructure of Ti-Si-C compounds at the SiC interface. Insufficient temperature leads to discontinuous reaction layers, while excessive temperature causes brittle phase overgrowth and microcracks. Optimal joining occurs at 850 °C, where the reaction layer is continuous and moderately thick (1.0 μm), simultaneously maximizing thermal conductivity ($21.6 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$) and shear strength (39.5 ± 1.0 MPa).

CRediT authorship contribution statement

Wendi Zhao: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Xiaoqing Si:** Writing – review & editing, Visualization, Supervision, Investigation, Data curation, Conceptualization. **Peng Li:** Writing – review & editing, Writing – original draft, Resources, Investigation, Data curation. **Yafeng Wang:** Writing – review & editing, Writing – original draft, Resources, Investigation, Data curation. **Bo Yang:** Supervision, Methodology, Investigation, Formal analysis. **Chun Li:** Writing – review & editing, Supervision, Methodology, Data curation. **Junlei Qi:** Writing – review & editing, Supervision, Formal analysis, Data curation. **Jian Cao:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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